

# **Project**

## **Advanced Surveillance Robocar**

Submitted By

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**For**

**L. Y. B.Tech Electronics Engineering**

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**2025-26**

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# Introduction

Modern military operations demand high levels of situational awareness, rapid decision-making, and effective surveillance in diverse and often hostile environments. Traditional surveillance methods expose personnel to significant risk, especially in combat zones or hazardous areas. To address these challenges, this project proposes the design and development of an Advanced Military Surveillance Robot equipped with multi-functional capabilities for reconnaissance, threat detection, and intelligence gathering.

The proposed robot builds upon the core concept of an omnidirectional IoT-enabled surveillance bot and incorporates advanced features such as nerve gas detection, autonomous navigation, landmine detection, night vision, AI integration, and bunker detection. This will enable defense forces to remotely monitor, detect threats, and act proactively while minimizing human exposure to dangerous conditions.

## Literature Overview

Modern military operations demand high levels of situational awareness, rapid decision-making, and effective surveillance in diverse and often hostile environments. Traditional surveillance methods expose personnel to significant risk, especially in combat zones or hazardous areas. To address these challenges, this project proposes the design and development of an Advanced Military Surveillance Robot equipped with multi-functional capabilities for reconnaissance, threat detection, and intelligence gathering.

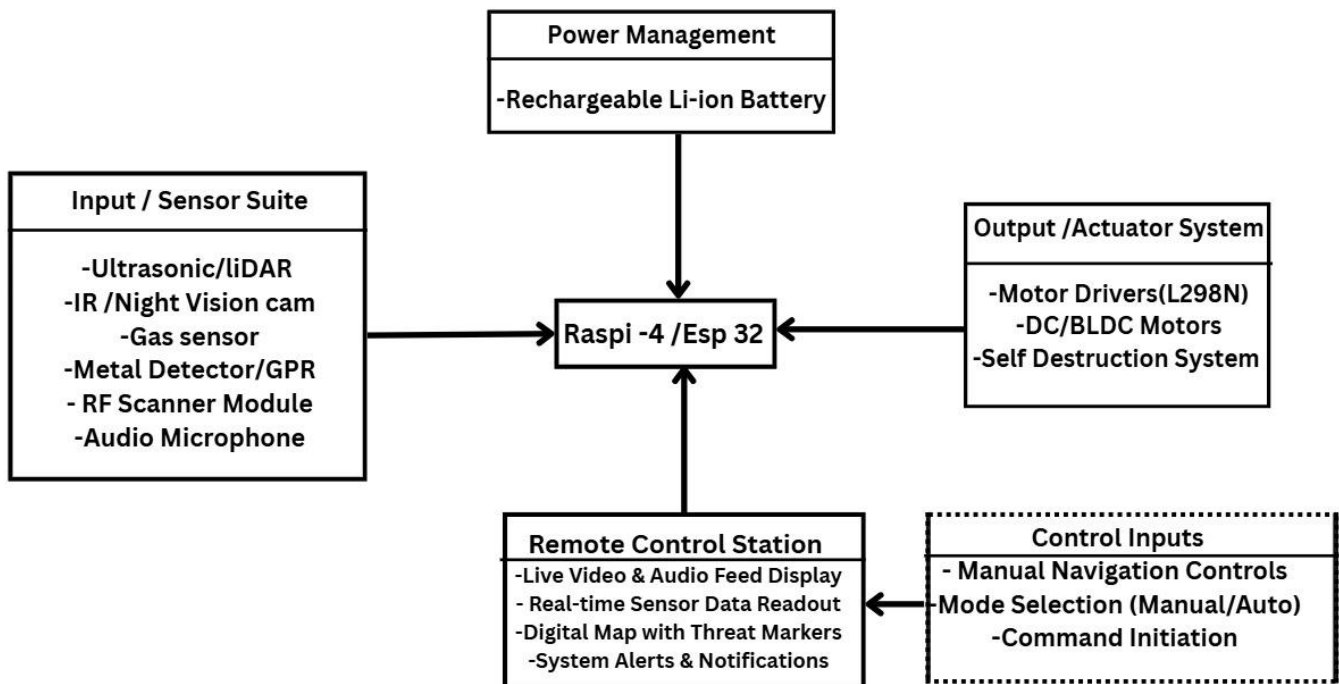
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# Project Overview

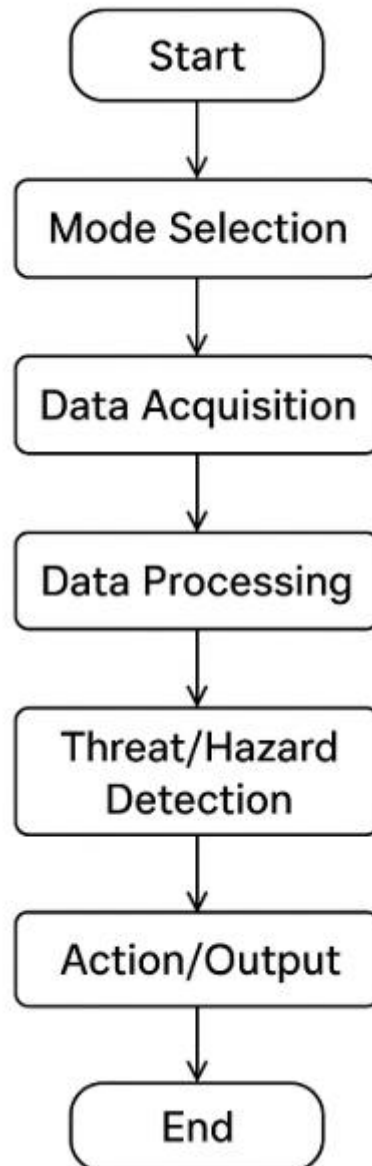
The main objectives of the Advanced Military Surveillance Robot are:

- Remote Surveillance – Provide live video, audio, and sensor data to a control station via.
- Hazard Detection – Identify nerve gases, hazardous leaks, and underground landmines, unknown humans & suspicious activity.
- Autonomous Navigation – Implement obstacle detection and pathfinding to avoid collisions along with manual control.
- Threat Response – Integrate an alert system for suspicious activities and a self-destruction mechanism for security in hostile captures.
- Night vision and low-light operative device getting rid of season and time constraints.
- Bunker Detection – Detect hidden enemy fortifications.

## Block Diagram



## Flow Chart



## Working

The robot will operate in both manual and autonomous modes:

In manual mode, operators control the bot remotely via a control station interface, receiving live video/audio feeds and sensor readings.

In autonomous mode, the robot uses obstacle detection sensors and pathfinding algorithms to navigate independently.

The gas sensor detects harmful nerve gases, triggering immediate alerts.

The metal detection module scans for buried landmines, marking them on a digital map.

Night vision cameras allow surveillance in darkness, while RF scanners detect enemy communication devices.

AI integration processes real-time data for threat classification and assists in decision-making.

## **Components Required**

- **Electronic Components :**

- Raspberry Pi 4 / NVIDIA Jetson Nano
- Omnidirectional wheel chassis with DC/BLDC motors
- Motor driver module (L298N or equivalent)
- Ultrasonic/LiDAR sensors
- Gas sensor (MQ-series or specialized nerve gas detector)
- Metal detector module or GPR kit
- IR/Night vision camera
- RF scanner module
- Rechargeable Li-ion battery with BMS
- Self-destruct mechanism (controlled fuse or physical destruction **circuit**)

- **Software:**

- Python / C++ programming
- OpenCV for image processing
- ROS (Robot Operating System) for autonomous navigation
- IoT cloud platform (AWS IoT, ThingsBoard, Blynk)
- GPT-based AI integration (via API)

## **Applications**

1. Border security surveillance.
2. Detection of chemical and explosive threats.
3. Battlefield reconnaissance.
4. Disaster zone inspection (collapsed buildings, hazardous spills).
5. Intelligence gathering in enemy territory.

## **Advantages**

- Reduces risk to human soldiers in dangerous missions.
- Multi-functional system integrates multiple defense capabilities.
- Operates in hazardous, low-light, and GPS-denied environments.
- Can be deployed for both defense and disaster rescue operations.

## **Disadvantages**

- High cost of advanced sensors and AI processing hardware.
- Limited operational time due to battery constraints.
- AI decisions require extensive training data to avoid false positives.